

Prathmesh Jain

+91 8657376595 | prathmeshjain42@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Engineer experienced in building web applications and backend systems using React, Node.js, and FastAPI. Skilled in designing APIs, distributed systems, and scalable applications with strong software engineering fundamentals.

EXPERIENCE

Tata Consultancy Services (TCS)

Oct 2024 – Present

Full Stack Engineer (Research & Innovation, CTO Team, BFSI Americas)

Mumbai, IN

- Built asynchronous workflow systems using **FastAPI, React, and LLM integrations** to automate compliance screening, document processing, and payment repair operations across multiple internal platforms.
- Developed concurrent backend pipelines for Google Custom Search API calls, web crawling, document extraction, and verification tasks, enabling long-running processes to execute independently while improving overall system responsiveness.
- Implemented agent-driven workflows using LLM-based orchestration for adverse media analysis, credit memo generation, and payment repair flows involving multi-step validation and tool-based reasoning.
- Designed real-time streaming interfaces using Server-Sent Events to stream workflow progress, intermediate outputs, and asynchronous task updates directly to frontend applications.
- Built high-throughput extraction and verification pipelines using Playwright-based crawling, multithreaded document processing, and LLM-assisted analysis across large unstructured datasets and financial reports.
- Developed persisted and resumable workflows with rollback handling, validation checkpoints, and human review stages to support long-running financial spread generation and transaction repair operations.
- Implemented reliability patterns including retries, fallbacks, circuit breakers, exponential backoff, and graceful degradation across AI-based systems to improve fault tolerance and user experience during partial failures.
- Worked across frontend and backend systems using React, FastAPI, PostgreSQL, Zustand/Redux, RBAC, authentication flows, file uploads, and Docker-based deployments while contributing as a core contributor across multiple projects.

PERSONAL PROJECTS

Patronix – E-commerce Platform | *React.js, Node.js, Express.js, MongoDB*

[Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack e-commerce platform with live hosting, secure authentication, product catalog management, cart, payment gateway integration and order tracking flows.
- Optimized application performance and global state handling using Redux Toolkit.
- Designed modular, maintainable RESTful APIs with token-based authentication and clean architecture.

ExpenseGauge – Expense Tracking App | *React Native, Node.js, MongoDB*

[Live Demo](#) | [GitHub](#)

- Designed and shipped a cross-platform, offline-first expense management application with real users and a downloadable Android APK.
- Implemented optimistic UI updates and background sync queues to ensure seamless usage during network outages.
- Implemented secure authentication using short-lived access tokens and rotating refresh tokens for seamless session continuity.
- Developed admin dashboards for user management, transaction audits, and balance adjustments.
- Built role-based backend services supporting expense analytics with monthly trends and category-wise visualizations.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

Backend: Node.js, Express.js, FastAPI, Flask

Frontend: React.js, Next.js, React Native, Redux Toolkit, Zustand, TailwindCSS

Database: MongoDB, PostgreSQL

Tools: Docker, Git, Postman, Playwright

EDUCATION

Vidyalankar Institute of Technology

Mumbai, IN

Bachelors in Computer Engineering

May 2024

GPA: 9.2 / 10